

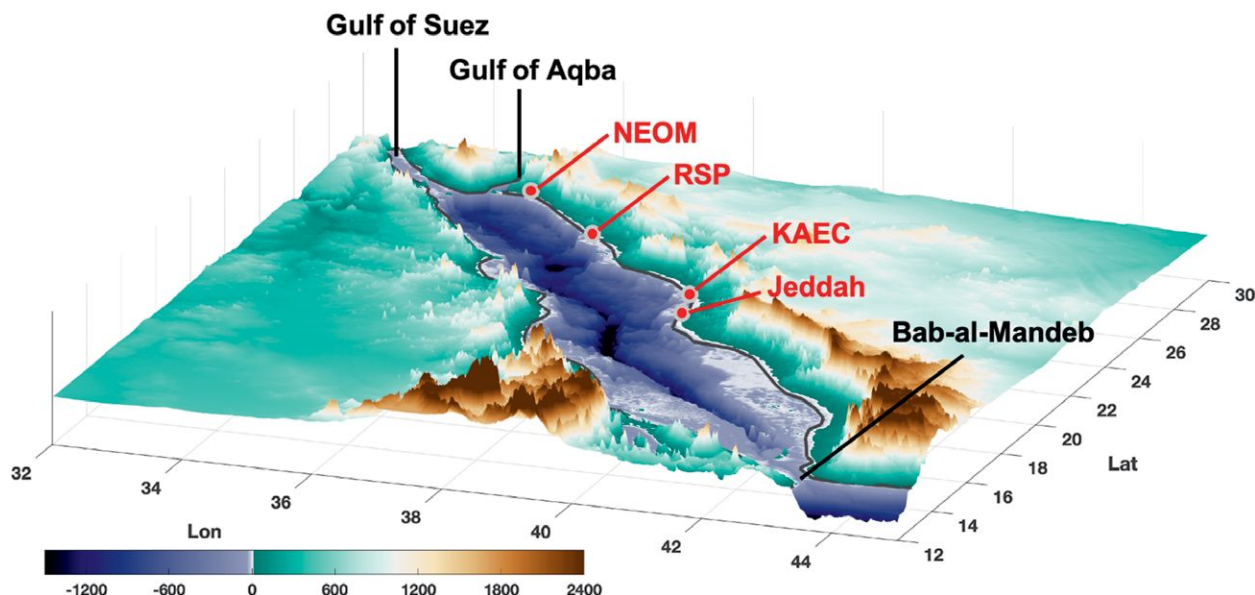
Modeling and Monitoring Weather and Climate Characteristics of the Red Sea Region

Adapted from "Towards an End-to-End Analysis and Prediction System for Weather, Climate, and Marine Applications in the Red Sea," by **Ibrahim Hoteit** (King Abdullah University of Science and Technology), **Yasser Abualnaja**, **Shehzad Afzal**, **Boujemaa Ait-El-Fquih**, **Triantaphyllos Akylas**, **Charls Antony**, **Clint Dawson**, **Khaled Asfahani**, **Robert J. Brewin**, **Luigi Cavaleri**, **Ivana Ceroveckí**, **Bruce Cornuelle**, **Srinivas Desamsetti**, **Raju Attada**, **Hari Dasari**, **Jose Sanchez-Garrido**, **Lily Genevier**, **Mohamad El Gharamti**, **John A. Gittings**, **Elamurugu Gokul**, **Ganesh Gopalakrishnan**, **Daquan Guo**, **Bilel Hadri**, **Markus Hadwiger**, **Mohammed Abed Hammoud**, **Myrl Hendershott**, **Mohamad Hittawe**, **Ashok Karumuri**, **Omar Knio**, **Armin Köhl**, **Samuel Kortas**, **George Krokos**, **Ravi Kunchala**, **Leila Issa**, **Issam Lakkis**, **Sabique Langodan**, **Pierre Lermusiaux**, **Thang Luong**, **Jingyi Ma**, **Olivier Le Maitre**, **Matthew Mazloff**, **Samah El Mohtar**, **Vassilis P. Papadopoulos**, **Trevor Platt**,* **Larry Pratt**, **Naila Raboudi**, **Marie-Fanny Racault**, **Dionysios E. Raitsos**, **Shanas Razak**, **Sivareddy Sanikommu**, **Shubha Sathyendranath**, **Sarantis Sofianos**, **Aneesh Subramanian**, **Rui Sun**, **Edriss Titi**, **Habib Toye**, **George Triantafyllou**, **Kostas Tsiaras**, **Panagiotis Vasou**, **Yesubabu Viswanadhapalli**, **Yixin Wang**, **Fengchao Yao**, **Peng Zhan**, and **George Zodiatis**. Published online in *BAMS*, January 2021. For the full, citable article, see [DOI:10.1175/BAMS-D-19-0005.1](https://doi.org/10.1175/BAMS-D-19-0005.1).

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The Red Sea, home to the second-longest coral reef system in the world, is a vital resource for the Kingdom of Saudi Arabia. The Red Sea provides 90% of the Kingdom's potable water by desalinization (it is one of the world's saltiest water masses), and it supports tourism, shipping, aquaculture, and fishing industries, which together contribute about 10%–20% of the country's GDP. These activities, and those elsewhere in the Red Sea region, critically depend on the regional oceanic and atmospheric conditions.

All major oceanographic processes occur in the Red Sea, but at much smaller spatiotemporal scales compared to the global ocean. This makes it an ideal test basin for studies of climate impact on regional ocean circulations, as the Red Sea is expected to exhibit a faster response to global climate variations



Atmospheric conditions over the Red Sea are mainly influenced by the arid surrounding terrestrial environment, the seasonal Indian monsoon system, and the orographic effects of the coastal mountains, with the resultant atmospheric forcing exhibiting marked variability on various time scales.

The last decade has witnessed an unprecedented increase in population and an acceleration in residential, commercial, and industrial

Topography of the Red Sea region. The important sites along the Saudi coastline are indicated on the map.

developments along the Red Sea coast. NEOM, short for the Latin Arabic term Neo-Mustaqbal (see table of acronyms), signifying “new future,” is a \$500 billion mega-project that aims at building a fully automated city operating as an independent economic zone in the north-western part of Saudi Arabia. The Red Sea Project (RSP) is another mega-development along the central Saudi coast, intended to transform the Red Sea coast into a world-class tourist destination.

Maintaining a sustainable, healthy marine environment is central to these developments and a core objective of Saudi authorities. In this context, understanding and forecasting the regional weather through climate variations is essential for a wide variety of high-impact societal needs, including environmental protection, coastal management, and mitigating weather extremes and pollution/contamination.

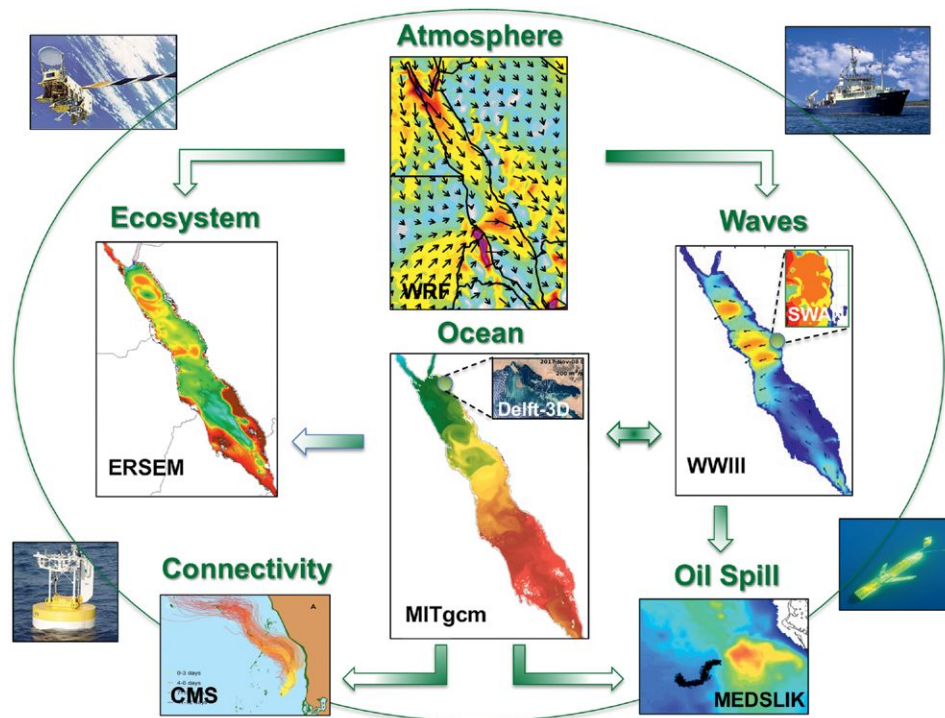
Toward this end, the Red Sea Modeling and Prediction group at King Abdullah University of Science and Technology (KAUST) has led the development of an integrated data-driven regional coupled modeling system [the integrated Red Sea Model systems (iReds)] to study, monitor, and predict the atmospheric and oceanic phenomena of the region.

In its present form, the WRF model is used to force various ocean and transport models. The regional ocean and wave models, based on the MITgcm and WWIII, were further recently fully coupled with WRF. Regional and

Acronyms and Abbreviations

Acronym/abbreviation	Expansion
Chl-a	Chlorophyll-a
CMS	Connectivity modeling system
Delft-3D	Deltares hydrodynamic flow model
ERA-5	ECMWF Re-Analysis, Version 5
ERSEM	European Regional Seas Ecosystem Model
iReds	Integrated data-driven Red Sea Modeling system
KAEC	King Abdullah Economic City
KAUST	King Abdullah University of Science and Technology
MEDSLIK	Oil spill, trajectory, and pollutant transport prediction model
MITgcm	Massachusetts Institute of Technology general circulation model
NEOM	Neo-Mustaqbal
RSP	Red Sea Project
SKRIPS	Scripps–KAUST Regional Integrated Prediction System
SWAN	Simulating Waves Nearshore
WRF	Weather Research and Forecasting
WWIII	WaveWatch-III

coastal configurations simulate and predict the general and meso-scale circulation, and 4-km to 100-m ecosystem models simulate the biogeochemistry of the basin. In addition, complementary (probabilistic) transport models are used to predict dispersion of contaminant plumes, oil spills, and marine ecosystem connectivity. Advanced ensemble data assimilation capabilities, incorporating in situ and satellite remotely sensed observations, have also been implemented to enhance forecasting skill.

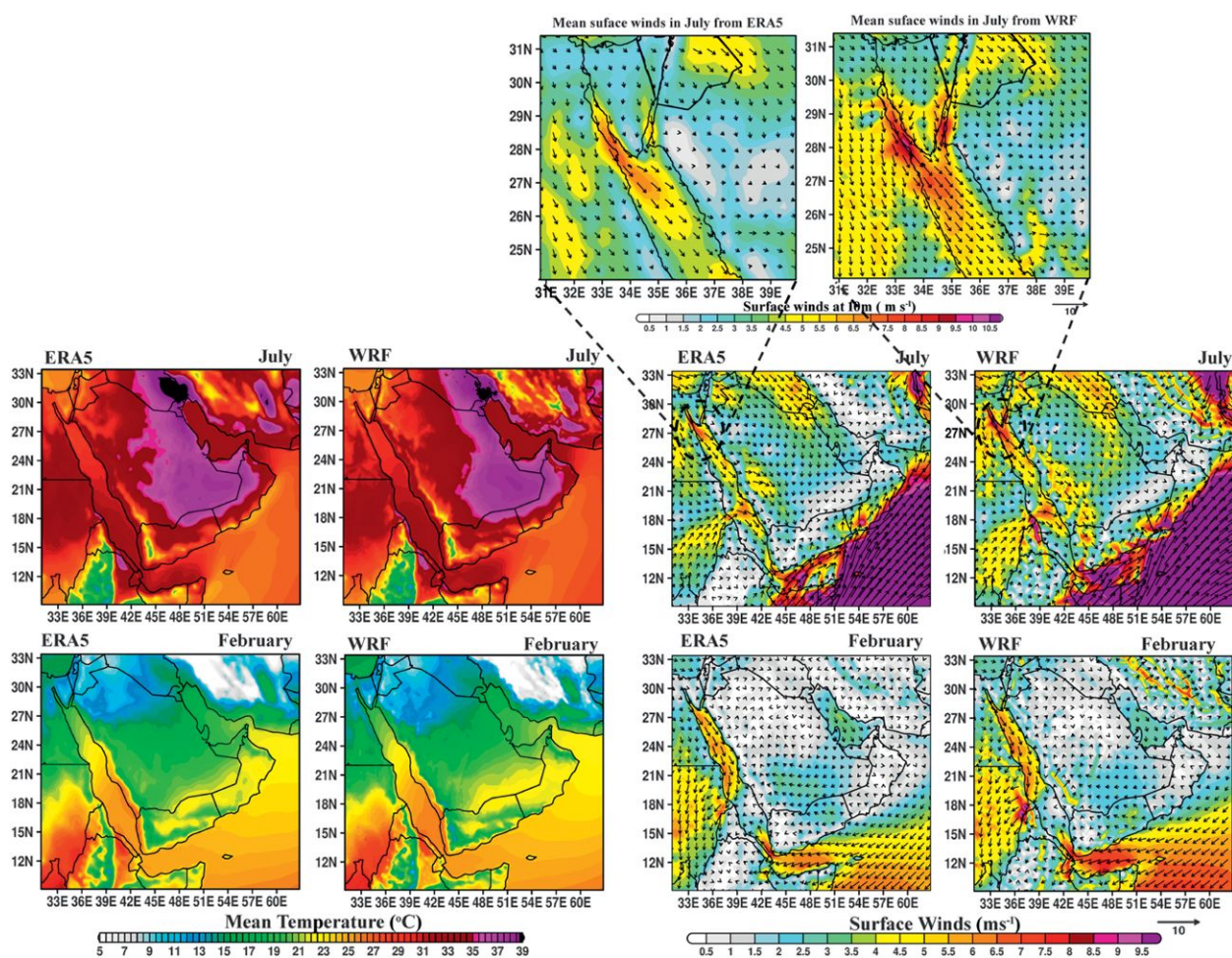


iReds accomplishments so far

In the span of just a decade, iReds has contributed new insights into the regional atmospheric and oceanic circulations, their variability, and trends in the Red Sea basin. Outputs from its various long-term simulations have revealed, in particular, unique basin-wide eddy activity and its driving forces, and a prominent seasonally varying overturning circulation modulated by the Indian monsoon. The system was also critically important in supporting the ongoing mega-projects along the Red Sea shores, providing unique information about the regional environment to optimize the design and operations of these new developments. It further developed, for the first time, regionally tuned forecasting capabilities successfully tested during urgent environmental situations. In addition, this effort has produced the first high-resolution atmospheric reanalysis for the region from 1980 to the present, generated using our assimilative atmospheric modeling system, critical not only for the Red Sea climate research, but also contributing to Indian monsoon studies. This regional reanalysis is being used to force the various ocean and transport models tuned to the Red Sea environment. Notably, these capabilities currently support various industrial and governmental developments.

▲ Schematic illustration of the data-driven Red Sea modeling and prediction system (iReds). Two-way arrows signify two-way coupling, and one-way arrows represent forcing of uncoupled components. The atmosphere panel shows mean wind speeds during the month of July from the 5-km WRF Red Sea reanalysis. The ocean panel shows a sea surface temperature snapshot in September with the upper-corner subpanel displaying the volume transport in the upper layer of a high-resolution coastal nesting exercise in the NEOM region. The ecosystem panel shows the simulated near-surface chl-a during the winter-mixing period (January–March). The waves panel shows a significant wave heights and mean wave directions snapshot in January with the upper-corner subpanel displaying the distribution of maximum significant wave heights estimated from a high-resolution nested coastal model long run covering the KAEC region. The connectivity and oil spill panels show simulations of oil and virtual larvae trajectories released in the central Red Sea.

Advanced visualization is an integral component of iReds to support data analysis tasks for scientists and nonexpert users. We have developed a visual analytics application, called RedSeaAtlas, based on the tasks and computational and operational requirements



of the variety of simulation models and observational datasets of the Red Sea region as part of the system.

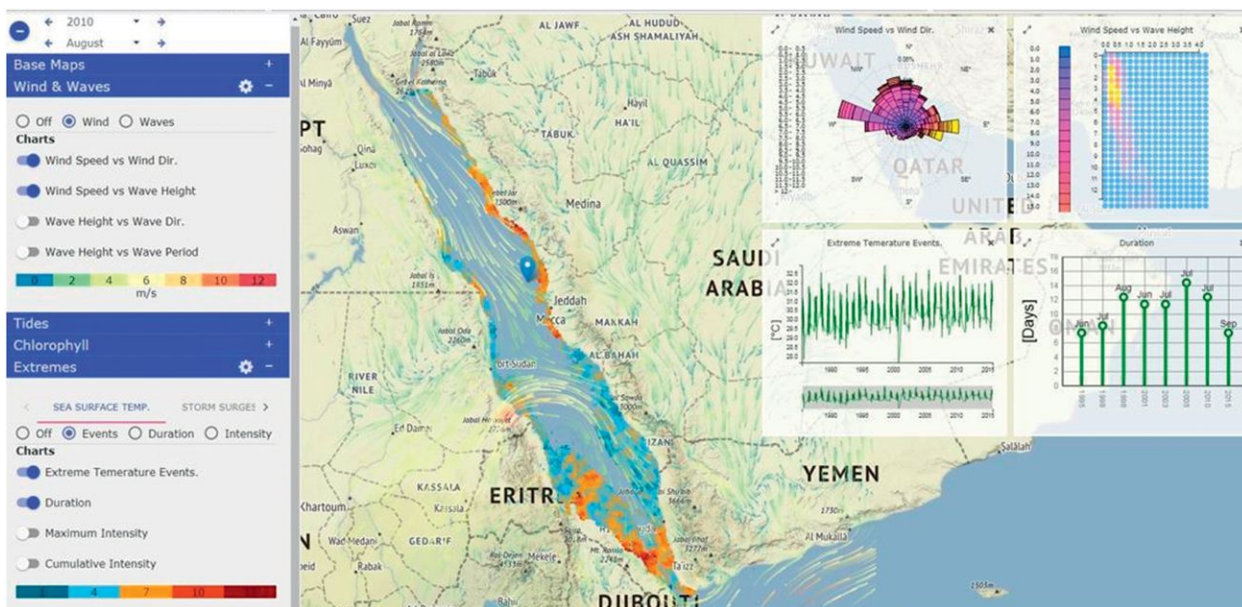
Future directions

We have begun work on implementing new capabilities to enhance iReds performances in the future, while further deepening our understanding of the regional circulation and variability. Our ultimate goal is to develop an integrated data-driven modeling and forecasting system that incorporates all available oceanic and atmospheric observations in the Red Sea region and makes predictions and actionable information available to scientists, government and industry, and nonexpert recipients.

In this regard, the first Red Sea operational system is already online, being tested for official release this year to provide weekly forecasts of the basin-wide atmospheric (WRF)–oceanic (MITgcm)–wave (WWIII) conditions at

Figure 4. The spatial distribution of (middle) summer (July) and (bottom) winter (February) months climatology of air temperature at 2-m height, and winds at 10-m height from the 5-km Red Sea reanalysis compared against ERA5. The Red Sea reanalysis reproduces well the main regional wind features in summer and winter, in good agreement with ERA5. (top) Zoomed-in views over the northern Red Sea region to emphasize the wind features captured by the high-resolution Red Sea reanalysis winds.

4-, 2-, and 1-km resolutions, respectively. It will be subsequently equipped with state-of-the-art interactive visualization and data analytics to provide decision-makers with real-time access to relevant information. Future work will then focus on (i) equipping iReds with new capabilities for forecasting aerosols (in particular, their affect on dust and pollution, the radiation budget, and clouds and precipitation)



and the marine biogeochemistry of the region, and (ii) extending the forecasting time range of the system to subseasonal-to-seasonal and annual scales through a combination of coupled modeling and data assimilation, and the spatial range to urban and coastal scales through downscaling.

So far, aerosol variations have only been climatologically accounted for in iReds. The WRF is equipped with state-of-the-art modules to simulate and study aerosols (WRF-Chem), and we have recently implemented it over the Arabian Peninsula. We are currently working on developing high-resolution aerosol emission inventories (natural, anthropogenic, and biogenic) by tuning satellite remote sensing retrieval algorithms with available in situ measurements. These will be incorporated/assimilated into WRF-Chem to simulate and predict dust storms and outbreaks of industrial and urban haze pollution.

Work is already underway at KAUST along with external collaborators at Imperial College (United Kingdom) and Deltares Institute (Netherlands) to build the required computational fluid dynamics algorithms to downscale our models' outputs and capabilities at urban scales. The plan is to investigate novel approaches for identifying efficient downscaling of probabilistic maps through machine and dynamical learning approaches.

We plan to continue working on enhancing the skills of our ensemble system. In particular,

The RedSeaAtlas visualization system showing overall wind patterns and statistics (at the dropped pin location, here north of Jeddah in the central Red Sea) of extreme temperature events in the Red Sea. Users can select any region to show more detailed information and associated attributes through different charts.

proper accounting of model errors through stochastic perturbations and forcing ensembles greatly enhanced the assimilation results. Developing data-driven approaches for quantifying the uncertainties of such perturbations and eventually estimating their parameters as part of the assimilation process is our next target.

Remote sensing will continue to be an integral component of the system in this historically data-sparse region, providing information for model validation and assimilation, and for predicting how the ecosystem indicators would respond to a changing climate. Monitoring the decadal and seasonal variability of Red Sea ecosystem indicators, in relation to the regional environmental conditions and broader climate change, is a necessity for the sustainability of marine life and responsible stewardship of its socioeconomical assets.

Finally, we will assess the system predictability of regional phenomena at different scales, including extreme weather events (heavy rainfall, air and marine heatwaves). The focus will be on identifying key factors and dynamical processes that may provide extended predictability for the region and quantifying their impact on forecasting skills through sensitivity studies. These will be then used as a guide for designing efficient, regionally tuned, data-driven strategies for projecting the weather and climate of the Red Sea and

associated impacts on the regional ecosystem and Red Sea coast.

We will then intensify studies using the newly developed Scripps–KAUST Regional Integrated Prediction System (SKRIPS) to understand coupled phenomena in the Red Sea region at short- and long-term scales, and how those respond to the global climate dynamics and variations. We are currently examining the outputs of the ocean-waves coupling component to investigate wave current interactions and coastal

circulations, with particular interest in the water exchanges near and over the coral reefs. Another proposed development of SKRIPS is the coupling of the atmospheric chemical and marine ecosystem components to investigate regional carbon variations and exchanges.

This coupled system will be at the core of building long-term regional predictive capabilities at the subseasonal-to-seasonal through climate scales by downscaling global forecasts and climate change scenarios. ●●

≡ METADATA

BAMS: What would you like readers to learn from this article?

Ibrahim Hoteit (King Abdullah University of Science and Technology): The Red Sea has long been understudied. I would like readers to learn about our ongoing research activities in the region, including the unique, integrated data-driven modeling system we are developing, our main scientific findings about the regional circulation and climate and their impact on the ecosystem, our future directions, and also our international team.

BAMS: How did you become interested in the topic of this article?

IH: Very little was known about the Red Sea circulation when we arrived at KAUST, which hosts a unique ecosystem fringed with coral reefs. It became my backyard, and there was a lot of interest in it, not only scientifically but also from the government to protect its valuable ecosystem. It was also part of the KAUST strategic vision to support these scientific and governmental activities. I moved to KAUST from the Scripps Institution of Oceanography, where I was working on modeling and studying the prediction of ocean general circulations in other marine environments. It

was quite natural to start working on the Red Sea. The more I learned about it, the more interested I became in understanding it.

BAMS: What surprised you the most about the work you document in this article?

IH: The Red Sea mesoscale eddy variability; the role of the Indian monsoon in seasonally reversing its overturning circulation and how this explains, for example, the surprising summer chlorophyll blooms in the southern Red Sea; and also the occasional deep-water formation events and the role of the faraway volcanoes in triggering the largest of these events in 1983 and 1992, etc. Another surprising finding is the high likelihood of decreasing Red Sea temperatures in the coming years.

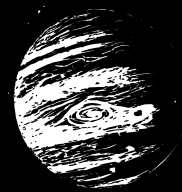
BAMS: What was the biggest challenge you encountered while doing this work?

IH: One of the biggest challenges when we started was the scarcity of the data in this basin. We used advanced numerical modeling and remote sensing to compensate for the lack of in situ data, and also to be able to go backward in time and reconstruct the history of the Red Sea circulation. This enabled us to

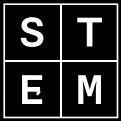
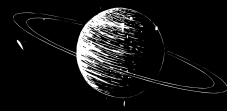
study and understand its variability at different spatiotemporal scales in a way that had not been possible before.

Another challenge we had to overcome was the lack of reliable high-resolution atmospheric fields to support studies of the ocean circulation and to force our ocean models. This is why we embarked on a subproject to reconstruct the history of the regional atmospheric circulation at high resolution to resolve the scales of the Red Sea region.

Other challenges included the need for enormous computational resources and the management of very large datasets, and building (from scratch after moving to KAUST) and managing a large multidisciplinary team with various expertise in ocean modeling, physical oceanography, atmospheric modeling, remote sensing, marine ecosystems, high-performance computing, applied mathematics, visualization, etc. Fortunately, KAUST played a major role in providing the facilities and resources necessary to conduct this research. Saudi ARAMCO (the Saudi Arabian Oil Company), which sponsored this work, also played an important supporting role by providing valuable data, without which we would not have considered expanding the scale of our work in the way we did.



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